

FAN SPEED INCREASING REGULATOR

Using PIC16F73

PROF. K. PADMANABHAN

Voltage dips in mains supply make fans run very slow, leaving you sweaty through hot summers, especially if you don't own an air-conditioner (AC). Voltage may also dip because of an AC drawing a large current, causing the fan to slow down.

The so-called fan regulators fitted on the switchboard for speed control merely reduce the fan speed, not increase it. Earlier, regulators used to have resistors that got connected in series with the fan to drop the voltage and hence the speed. These had a rotating switch with steps 1 to 5 to set the speed below normal, and were fitted on the panel board along with the fan switch, light switch and other power outlet sockets.

Nowadays regulators use triac to slice the AC waveform and thereby provide an effective drop in voltage and hence speed. These regulators are small in size and can be fitted along with switches compactly.

At present, there is no fan regulator which can actually increase the fan speed beyond its maximum when required. To increase the fan speed, you can increase the voltage, but that is not feasible. Connecting an auto-transformer with boost voltage increases the speed somewhat, but that is not worth the effort.

Usually, fans run at maximum speeds of around 300rpm, but at low voltages, the speed drops by more than 35 per cent. Ceiling fan motors have about 18 poles or more. The full or synchronous speed of the fan motor is given by:

$$120 / f p$$

where 'f' is the mains supply frequency and 'p' the number of poles. The running speed is much less than this value.

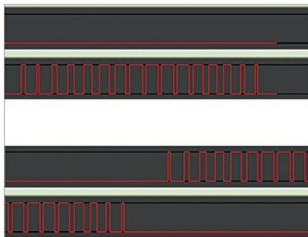


Fig. 1: Pulse width changes as per the sine waveform

Thus a fan's maximum speed can be increased only by increasing the frequency of the alternating current supplied to the fan. Increasing this frequency to 60Hz or even 65Hz will make the fan run much faster and produce a good air throw.

If a 330rpm fan is supplied with only 70 per cent of the mains voltage due to a voltage dip, the torque of the motor reduces to half its full-voltage torque value. The torque needed to drive the fan blades is proportional to the square of the speed. So the speed drops to 70 per cent when the torque drops to half its full value.

But, if you increase the frequency to 60Hz with the same reduced voltage, the speed increases 20 per cent due to the increase in frequency from 50 to 60 Hz, which makes the speed drop by only 10 per cent of the nominal 330rpm. If you increase the frequency to 65 per cent, the fan will run even faster.

Increase in frequency does not affect the fan's motor much. Actually, at the increased frequency, the magnetic flux density in the iron of

the motor reduces, and that is good for efficient working of the fan's motor. The magnetic power loss does not increase much.

Thus, a method is needed to provide variable-frequency mains voltage derived power for the fan. This power has to be generated in a compact, economical and simple manner as possible.

There are power modules available for varying the speed of induction motors. These use insulated-gate bipolar transistors and are quite expensive.

PWM-based variable-frequency power

The best method for generating variable frequency in the range of 45 to 65 Hz is pulse-width modulation (PWM). Pulse widths change as per the sine waveform shown in Fig. 1. The pulse sequence should be generated by the microcontroller. In this project, the pulses (PWM1 and PWM2) alternately appear on two pins of the PIC microcontroller (refer Fig. 5). The timing of the pulses will be constant for a given frequency. The pulse sequence timing determines the frequency.

When pulses of current pass through the fan's inductive winding, the current averages to sinusoidal shape. Pulse discontinuities get filtered by the fan inductance. To generate the sine wave, two such pulse trains are required, one for the